

Article

Investigating the Impact of Perceived Micro-Level Neighborhood Characteristics on Housing Prices in Shanghai

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Abstract: It is widely accepted that houses in better-designed neighborhoods are found to enjoy a price premium. Prior studies have mainly examined the impact of macro-level neighborhood attributes (e.g., park accessibility using land use data) on housing prices. More recently, research has investigated the micro-level features using street view imagery (SVI) data, though scholars limited the scope to objective indicators such as the green view index and sky view index. The role of subjectively measured street qualities is less discussed due to the lack of large-scale perception data. To provide better explanations of whether and how the micro-level neighborhood environment affects housing prices, this article introduces a framework to collect designers' perceptions on five subjective urban design perceptions from pairwise SVI rankings in Shanghai with an online visual survey and further predicted through machine learning (ML) algorithms. We also extracted ten important objective features from the scenes. The predictive power of micro-level neighborhood street perceptions (subjective perceptions and objective features) on housing prices was investigated using the hedonic price model (HPM) through ordinary least squares (OLS) and spatial regression, which considers spatial dependence. The findings prove the significance of the value of perceived qualities of the neighborhoods. It reveals that both objective perceived features and subjective perceptions significantly contribute to housing prices; while the objective features show more collective strengths, individual subjective perceptions have more explanatory power, and we argue that these two measures can complement each other. This study provides an important reference for decision makers when selecting street quality indicators to inform city planning, urban design, and community and housing development plans.

Keywords: street environment; micro level; human perception; housing price; street view imagery; computer vision; machine learning



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1. Introduction

1.1. Neighborhood Built Environment and Perceptions

The global pandemic (COVID-19) has tremendously changed how people live and work in a rather drastic manner. Many offices and companies proposed the hybrid home-office work mode or even decided to permanently transition to remote working [1,2]. In this context, people inevitably stay longer at home, which implies sedentary behavior [3]. The commuting to work mobility pattern was also completely altered because of various regulations; the radius of people's daily activities has primarily been confined to their neighborhoods where their residences are located. Thus, the built environment at the neighborhood scale has become paramount in the 'post-pandemic' era and has received much research interest. As a critical component of the neighborhood environment public realm,

streets provide important physical spaces for people to gather, connect, and interact [4]. The social benefits it brings to the neighborhood are critical, as it enhances the residents' social connectedness, protects people from isolation and loneliness, and enhances their mental well-being [5]. Researchers and city planners have increasingly realized the unique value of creating a more vibrant neighborhood. It is associated with quality of life, further improving the city's resilience [6].

Classical urban placemaking theory states that successful urban places have three essential aspects: physical space, sensorial experience, and activity [7]. These three parts are intertwined and overlapping. The physical space of a place provides an actual medium to facilitate activities, while the built environment stimulates the psychological elements which form part of people's perception of the place. Scholars have further noted that this human-environment interaction can provide opportunities to create a desirable neighborhood setting generating a sense of place [8]. At the neighborhood environment scale, the visual appearances of streets can affect perceptions and further influence behaviors. For example, higher buildings typically provide better visual enclosure, and a lower sky view factor [9] is generally perceived as safer. Moreover, street tree canopies can provide necessary shade and serve as a wind buffer to create thermally comfortable spaces for pedestrians, further improving microclimate control and mitigating the urban heat island effect. It is suggested that a higher visual greenery ratio in street canyons could lead to better-perceived greenery and aesthetic appreciation of the physical setting [10]. A following related study detected the positive correlation between perceived greenery and the odds of walking along streets [11]. These micro-level perceptions are reported to be associated with a myriad of attributes including the crime density [12], walkability [13], cycling and walking behavior [11,14], pedestrian walking and running route choice [15,16], mental and physical health [17], people's sensory experiences [18], and living quality [19]. When aggregated together, these individual perceptions may cause further issues for the neighborhoods, for instance, physical disorders like broken windows are more likely to attract crimes [20] and visual safety is correlated with income [21]. Therefore, micro-level perceptions of the neighborhood environment not only pose significant impacts on behaviors but also have implied socioeconomic outcomes such as on housing prices [21–24]. Thus, ignoring the perceptual and experiential qualities of street characteristics underestimates the value of human-centric design, which may cause biased suggestions and downplay the complexity of how people interact with the environment and the benefits that neighborhoods can bring in the long run.

1.2. Housing Prices, HPM, Macro- and Micro-Level Variables

As a necessary living commodity, housing is strongly influenced by the physical and socioeconomic environment. The research on house prices receives lasting attention and is viewed as a barometer of human settlement [22]. Its value is jointly determined by multifaceted factors, and it is a good reflection of people's direct preferences. In this regard, HPM is a widely adopted approach in housing price urban studies [25]. The hypothesis is that the house values are determined jointly by different attribute groups, including structural, location, and neighborhood attributes. These extensively studied indicators can be viewed as macro-level variables, referred to in this paper as objective characteristics collected at the macro scale, such as land use pattern and building density [15]. For example, the structural attribute group includes the time when the house was built, property area size, etc. [26,27]. The physical locations are typically represented by the location attributes, such as the relative locations of suburbs or urban peripheries [28]. Neighborhood attributes include the accessibility to important amenities (e.g., hospital, school, park, retail, etc.), distance to the central business district (CBD), and other factors [29,30]. Overall, this standard HPM has proved to be effective in previous empirical studies for urban planning and policymaking [31,32]. Nevertheless, previous studies reported that the spatial dependence of housing prices might result in inaccurate interpretations of coefficients, and the strength of the variables it revealed can be problematic when solely depending on the

OLS model for the HPM. If the residuals of the OLS model are spatially autocorrelated, the model is no longer valid [33]. We should consider expanding the research model by using a spatial regression model to improve the accuracy and technical soundness [34].

In prior studies, one drawback is that they completely ignored the micro-level perceived characteristics of the neighborhood street environment and their impact due to data availability. In contrast to the definition of ‘macro-level’, the micro-level variables represent how people read the visual streetscape in the street canyon and perceptions from the human-centric perspective. Scholars have argued that the determinants of a house buyer’s decision cannot be fully explained by the property and physical settings but also greatly depend on the unique social experience, how the neighborhood is perceived, and its vitality [22]. From this place-based perspective, how micro-level variables and people–environment interaction affect housing prices is under-addressed. Studies have tried to use objective indicators based on GIS data to proxy the street perceptions or quality, including tree canopy size [35]. More recently, SVI data have enabled the human-centric perspective for a street environment, and, through computer vision (CV) technology, scholars have demonstrated the positive association between housing prices and the street greenery [34,36], sky, and building view indices [37]. However, these studies only focused on single or very few objectively measured street features and so cannot fully capture the overall perception of a homebuyer.

Another major part which has been less discussed is the micro-level subjective perceptions. Subjective measures are derived from the perception surveys and questionnaires and have been integrated with ML algorithms to predict and map citywide perceptions in emerging urban analytic studies [23,38–41]. Kang et al. [22] studied how subjective perceptions (lively, safe, etc.) jointly affect housing prices. Nevertheless, these dimensions focused more on psychological aspects and are less concerned with the urban design perceptual qualities of the streets [42]. Tian et al. [41] extracted eight different subjectively measured landscape design qualities and compared the street characteristics of various zones in Berlin. Xu et al. [39] measured the street qualities in both subjective and objective measures for the same perception concepts, and the subjective measures were stronger in explaining housing prices. Song et al. [38] spatially mapped the pairwise perceptions using both measures in Shanghai and identified huge differences regarding the within-perception heterogeneity patterns when compared with their counterparts. However, to what extent the subjective perceptions complement the objectively measured street features is less investigated.

1.3. Research Questions, Research Gaps, and Contribution

What are the impacts of micro-level perceived characteristics measured using SVIs on housing prices in comparison to conventionally measured macro-level variables using HPM? How do objectively measured features complement or conflict with subjectively measured perceptions from SVIs? This study quantified how micro-level perceived characteristics affect property values, which is reflected by adding ten objective features and five subjective perceptions as separate attribute groups into the model.

In summary, we intend to address several knowledge gaps and the contribution of our research is four-fold. First, the existing HPM literature has relied on macro-level indicators, while the impact of fine-grained and micro-level neighborhood environment perceptions needs to be discussed more. We provide a high-throughput framework [38] that measures five subjective perceptions and ten key objectively perceived street features using SVI data. The study sheds light on whether micro-level neighborhood attributes influence housing prices compared to macro-level attributes.

Second, internally among neighborhood micro-level variables, the subjective urban design perceptions have long been under-addressed [23,39]. We hypothesize that the subjective perceptions can capture the overall feeling and experiential quality and may outperform the strength of the individual objective feature index [42]. This study provides interpretations on how each subjective design-oriented perception may contribute to housing premiums.

Third, studies have only explored the impact of limited objective street features, such as mainly focusing on greenery using HPM [34]. The coverage of other street features can be further expanded as these ubiquitous visual elements have been proven to be associated with subjective perceptions [40]. A deeper understanding of how they contribute to the price premium can enrich our knowledge and may provide interpretable implications.

Fourth, the subjectively measured perceptions may complement or contradict the objective street characteristics when examining their strengths in predicting property value; thus, their relationships can be further explored. This study also provides meaningful insights on variable selection for urban planning and policymaking decisions.

This research will provide a scientific framework for policymakers, planners, and multiple stakeholders to assess the perceived local environment and understand its economic implications on properties in the neighborhood in comparison to conventionally measured macro-level attributes. In the future, when formulating streetscape guidelines and associated urban planning policies, the tools employed in our research can be used to reflect in situ needs in a tangible way.

2. Data and Methods

2.1. Conceptual Framework

In summary, compared to well-understood macro-level factors, how and to what extent the micro-level human-perceived neighborhood environment affects housing prices needs further research [22]. These eye-level neighborhood attributes include both objectively measured street elements such as trees and subjectively measured perceptions which reflect the classical urban design qualities in previous empirical studies [42]. We hypothesize that objective features and subjective perceptions could have different impacts on the price premium. Thus, we need to understand which type has a more robust relationship.

This study aims to provide a framework that can effectively measure micro-level objective street features and subjective perceptions of the neighborhood street environment using SVI, crowdsourcing surveys, and CV combined with ML models. Our assumption is that micro-level factors have strong associations with housing prices and should complement traditionally measured macro-level factors, as micro-level factors are the proxies of how people perceive in daily life and directly affect people's moods and behavior, and hence the price premium. Among all the micro-level variables, the subjective perceptions may outperform the objective features to better explain the housing prices as they better reflect the comprehensive cognition process of people reading the visual landscape.

2.2. Analytical Framework

First, we downloaded citywide SVI data of Shanghai from the street view platform. Second, we collected five subjective perceptual qualities from an online visual survey. Third, we applied a semantic segmentation model to extract more than 30 key physical features. Fourth, the best-performed ML models were trained based on the visual survey results to predict the five subjective perceptions for the entire SVI dataset. Fifth, tree-based ML models can generate the Gini importance scores of each objective feature. We selected the ten most important elements as the objective street features to be further studied. Sixth, we added both subjective perceptions and objective physical features to the HPM as separate models using housing transaction data from the Lianjia website (secondary data from Qiu et al. [25]) and compared explanatory results from OLS and spatial regression models. In this process, accounting for spatial dependence, conventionally measured macro-level variables (including structural, locational, and neighborhood attributes) were used as control variables to understand which variables were more robust in explaining housing prices (Figure 1).

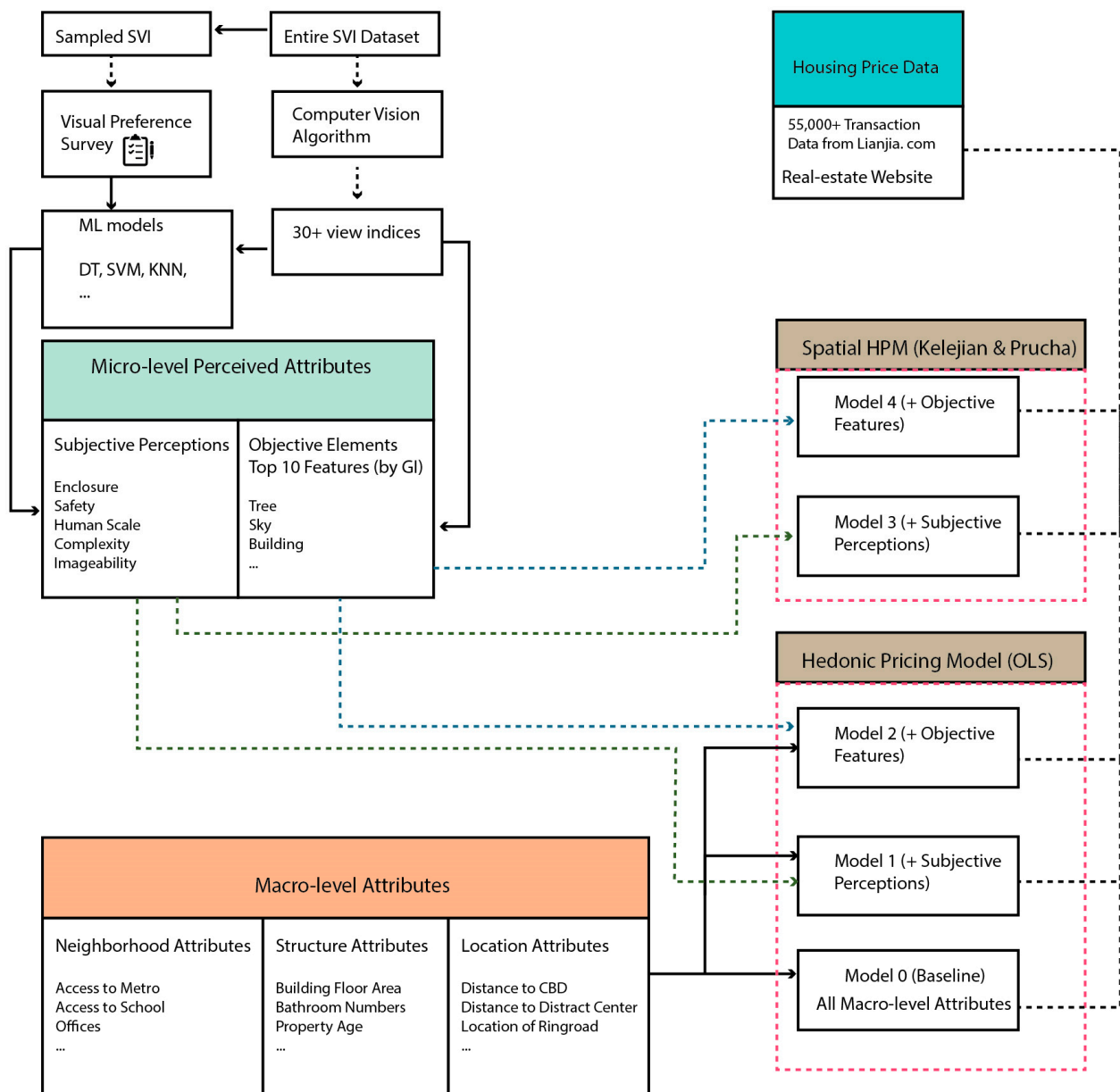


Figure 1. Methods and workflow.

2.3. Study Area

As one of the major financial, trade, and shipping hubs, Shanghai is a key real estate market in China [37]. A citywide neighborhood-level empirical analysis of the housing prices and neighborhood environment can reveal important implications for future housing policymaking, improve understanding of the desirability of a neighborhood, and provide meaningful suggestions to build a sustainable and resilient city.

2.4. Housing Transaction Data

Regarding the housing prices, using the year 2019 and Shanghai as criteria on the Lianjia.com (accessed on 20 August 2022) website (the primary Chinese real estate brokerage platform, secondary data from Qiu et al. [25]), we obtained transaction records of apartments. After cleaning the data of those records with (1) missing property attributes and (2) extreme prices (e.g., invalid value or extreme square meter price compared to mean value), we retrieved 40,159 valid records with coordinates (see Appendix A). The

average price of housing in Shanghai was 57,349 RMB/m² and the prices ranged from 10,400 RMB/m² to 250,813 RMB/m².

2.5. Selection of Micro-Scale Perceived Variables

We selected four subjectively measured qualities (i.e., enclosure, human scale, complexity, imageability) based on classical urban design theory by Ewing and Handy [42], together with the perception of safety [43]. These five qualities represented the subjective perceptions of the neighborhoods in Shanghai. Operationalized definitions of these qualities maintained a relatively good consistency in prior studies: Enclosure measures how the streets are visually enclosed by vertical features of the built environment including trees and buildings [9]; human scale represents the scale of a human at walking speed, for example, paving texture and street furniture; imageability proxies what makes the place different from others and memorable; complexity is related the richness of the visual perceptions of the streets and is associated with the variety of street features present in the scenes. Additionally, perceived safety is added because it tremendously affects human behavior [43].

Limited research has been conducted on the impact of eye-level objectively measured features on housing prices. The variables were confined to limited indicators such as visual greenery [34,36]. We argue that other types of objectively measured features can be further investigated to give a more comprehensive picture of their potential impact.

2.6. Calculating Micro-Scale Factors

2.6.1. Downloading SVI Data and Extracting Physical Features

SVI can represent the human-centric perspectives of pedestrians and becomes a great data source for urban studies [44]. We followed methods from previous studies [38,39] and sampled SVIs at 50 m intervals [45] along the road segments within 1 km radius of each property's coordinates in QGIS software, requested SVI data from Baidu Street View Static API, and retrieved 25,276 valid SVIs for our study. Previous studies indicate that neighborhood-level perceptions should be proxied in areas rather than points, and a 1 km radius is the approximate walking distance of a 15 min neighborhood, which is a widely suggested concept in urban design [13].

View index denotes the ratio of the feature's pixels to the entire scene and can be calculated effectively using a pyramid scene parsing network (PSPNet) algorithm [46], which has been very popular in emerging studies [38,47]. Specifically, the PSPNet is pre-trained on the ADE20K dataset which includes 150 object types tailored for cityscapes and can achieve around 80% accuracy [48]. Figure 2 shows examples of extracted features from the inputted SVI.

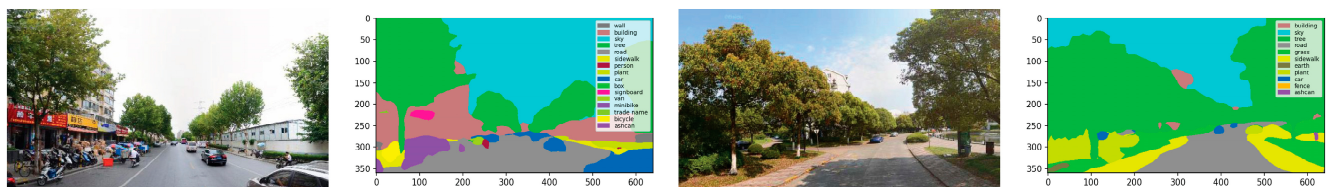


Figure 2. Example of CV parsing raw inputs and results.

The SVIs in total extracted 36 types of physical features with large variances in their quantities. Typical visual features at the street level such as sky and trees can have a direct impact on human perceptions [40,42] and have been tested by housing price research [37,49].

2.6.2. Collecting Subjective Perceptual Qualities and Key Objective Features

Locally collected subjective perception scores from expert panels can be used as training labels to predict citywide scale datasets using ML algorithms [39]. We randomly selected 300 SVIs across Shanghai from the entire SVI dataset covering different geographic

locations. We developed a visual survey platform to collect opinions from a panel of 43 participants (with architecture-related backgrounds so that they were well versed in urban design concepts). They were asked to choose a preferred image based on the definition of perceptions (Figure 3). To improve the score accuracy and reduce the bias, we adopted Microsoft TrueSkill algorithm to convert the collected pairwise preferences into interpretable scores [50]. The 300 SVIs were sufficient as training labels as we extracted roughly 30 features from SVI using PSPNet, and scholars have reported training samples need ten times of the number of variables to attain reasonable results [51].

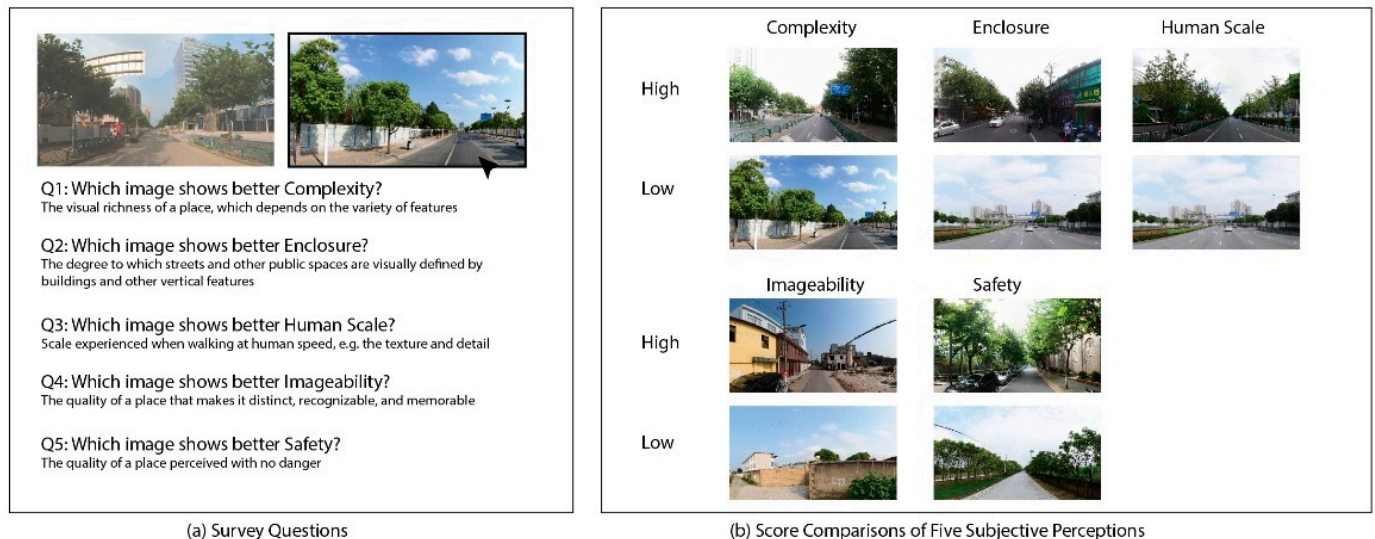


Figure 3. Crowdsourcing online survey to collect subjective opinions from panel of experts. (a) The pairwise images regarding five perceptions. (b) Pairwise comparison score examples.

The five subjective perception scores of the 300 SVIs were used as training labels, while the view index of each feature parsed from SVI served as explanatory variables. The 300 SVIs were split by 80% for training and 20% for testing. The tree-based models have been used in previous studies to predict urban scene perceptions because of their efficiency [52,53]. Since we have a similar task in our study, we adopted this method and applied multiple tree-based algorithms to predict the five perceptions. Preliminary prediction results (Table 1) were judged on R-squared (R^2), mean absolute error (MAE), and root mean square error (RMSE). We chose random forest (RF) based on the comparison results to apply to the entire SVI dataset for each perception. In general, given the small training sample size, the prediction accuracy is acceptable. Three out of five perceptions have the R^2 value ranging between 0.47 to 0.61, which can explain half of the variance, further raising the bar when compared with previous research results [42,54,55]. MAE indicates that enclosure and safety have the lowest value, implying that they might be more straightforward so they can reach greater consensus and lead to higher prediction accuracy [40].

Table 1. Prediction performance using RF algorithm.

Subjective Perceptions	R^2	MAE	RMSE
Complexity	0.49	0.13	0.16
Enclosure	0.61	0.09	0.11
Human Scale	0.51	0.12	0.14
Imageability	0.47	0.15	0.19
Safety	0.57	0.11	0.13

We further averaged these predicted subjective perception scores within 1 km of each property to represent the neighborhood-level street qualities [13]. Similar to previous research, subjective perceptions exhibit spatial clustering tendency and within-quality heterogeneity [38]. For instance, the west bank of Huangpu River displays higher scores than Pudong District as the west side is more established regarding amenities and density, while the new development in Pudong is relatively more dispersed.

Furthermore, to understand the visual features' relationship with subjective perceptions, Gini importance (GI) can be calculated to represent the importance of each explanatory variable [56]. We applied a tree-based regressor in the Python Scikit-learn package to calculate GI scores. We obtained the GI scores of street features that predict each subjective perception, summed them, and ranked them by value. We found large variances for each perception. The summed GI values further indicate that the sky and tree features are important, and a few other ubiquitous street elements rank higher too. For instance, sky view shows higher strengths in complexity than enclosure, and, surprisingly, the top several features previously reported as important features in implying perceived safety, including streetlights and street furniture, were not among the variables which ranked highest regarding GI [57]. We selected the elements ranked in the top 10 (i.e., sky, tree, building, person, car, road, sidewalk, fence, wall, and signboard) among all the extracted features (Figure 4) as variables to further fit into the HPM in the next step.

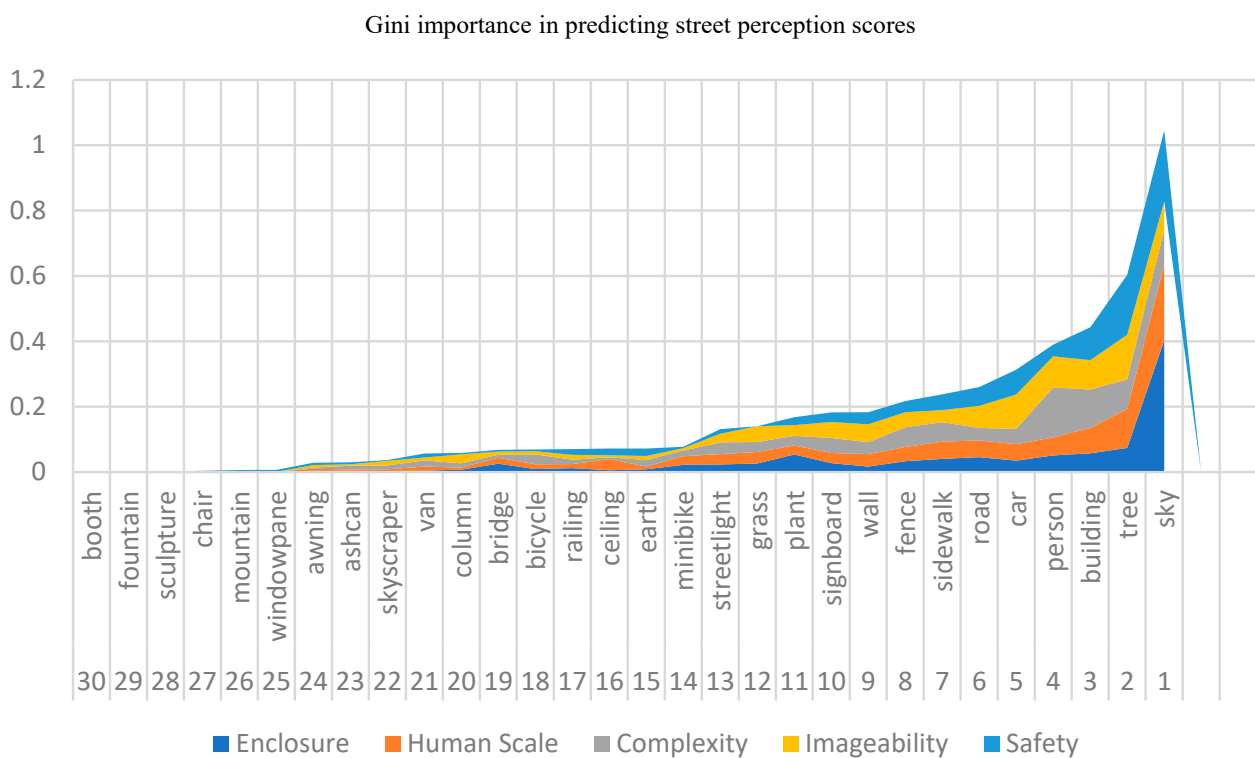


Figure 4. Gini importance of each objective feature of subjective perceptions' prediction.

2.6.3. Correlation Analysis for Perceptual Scores

Zhang et al. [40] reported high correlations in subjective qualities such as 'beautiful-wealthy' and 'depressing-safe', presenting multicollinearity issues. Therefore, Pearson correlation analysis was performed on micro-level urban design perceptions, and we checked the strength of the linear relationships between pairwise subjective perceptions as well as objective features. Notably, in our study regarding subjective perceptions, correlations between complexity–imageability and complexity–safety are low, while human scale–complexity shows a high positive correlation (0.82). This helps detect the multicollinearity when choosing explanatory variables for HPM. It suggests that more theoretical

efforts should be made to better define the concepts and reduce the ambiguities and overlaps. In terms of objective features, they mostly exhibit negligible or low associations, except for sky that has a moderate correlation with building. The following variance inflation factor (VIF) test confirmed its multicollinearity issue with other elements. Thus, those high-VIF features such as sky were removed.

3. Hedonic Price Models and Results

In the conventional HPM, the housing price is jointly dependent on three categories of attributes, i.e., structural, location, and neighborhood attributes [25]. To scrutinize the impact of micro-level variables, we added micro-level perceptions as a separate attribute category (STRE) but added each type of perceived attributes independently. Thus, in this study, we extend the HPM as:

$$\text{PRICE} = \alpha + \beta_1 \text{STRU} + \beta_2 \text{LOCA} + \beta_3 \text{NEIG} + \beta_4 \text{STRE} + \varepsilon \quad (1)$$

where α is the constant, β_1 to β_4 are the corresponding coefficients for structural (STRU), location (LOCA), neighborhood (NEIG), and streetscape (STRE) attributes, and ε is the error term.

However, the conventional OLS model assumes that the random errors have a normal distribution, thus causing biased implications if neighboring locations impact the place. The Moran's I test in the OLS residuals and the robust Lagrange multiplier (LM) tests can be applied to determine the spatial dependence in housing prices. The examination of our data exhibited both spatial lag and error terms. Therefore, the Kelejian and Prucha model was selected that contains both a spatially lagged dependent variable and autocorrelated error term [58]. The spatial HPM is as below:

$$\text{PRICE} = \rho W_{\ln \text{PRICE}} + \beta_1 \text{STRU} + \beta_2 \text{LOCA} + \beta_3 \text{NEIG} + \beta_4 \text{STRE} + \mu, \text{ where } \mu = \lambda W_{\varepsilon} + \varepsilon, \quad (2)$$

where ρ is a spatial autoregressive parameter, $W_{\ln \text{PRICE}}$ is a spatially lagged log price variable, W_{ε} is a spatially lagged error term, λ is the error lag parameter, and ε is a vector of residuals.

3.1. Dependent Variable—The Property Value

Housing transaction records were collected from [Lianjia.com](https://www.lianjia.com) (accessed on 20 August 2022) (secondary data from Qiu et al. [25]), providing price and structural and location information for the residences. After required data cleaning—to mitigate the spatial overlap issue as an average of 7.1 units per building was downloaded but each unit could have significant variances in structural characteristics (e.g., number of rooms)—we split the housing records into ten mutually exclusive random subsets [59] of which each contained 5000 unique apartments that avoid any spatial overlap. The housing price was log-transformed, and we ran regressions on each subset and these repeated efforts helped check whether the results were relatively stable [37,60]. The data description can be found in Appendix A.

3.2. Independent Variables—Macro- and Micro-Level Factors

Independent variables include both macro- and micro-level factors and were chosen based on literature and data availability (see Appendix A). The first type of macro-level attributes includes both continuous variables, such as construction year, and categorical variables (transformed into dummy variables) such as interior decoration quality. The second type, location groups, includes the physical distance to the (1) CBD and (2) nearest district center. Third, neighborhood attributes measure the distance and accessibility to amenities such as restaurants per km². For example, the living service density (DENS_{SRV}) calculated the total number of amenities, such as café and retail, grocery stores, and hospitals, per km² within the neighborhood's district boundary. In this category, we further included important amenities such as all metro stations and 68 high schools with

an excellent reputation for education which are recognized by the Shanghai government as providing high-quality education. Besides the closest distance measured using GIS network analysis, the accessibility was calculated using its numbers within a 1 km and 5 km radius of a property.

Among the micro-level factors, we used the five subjective perceptions for the regression, while the objective indicators have been studied before, including the sky, tree, and building view indexes, which were included based on a literature review [36,37,49]. We further expanded the indicators by integrating person, sidewalk, car, and fence which were studied by walkability research [45]. In conclusion, we selected ten objective features that have a modest to the most prominent existence in statistics and high Gini importance scores.

3.3. Model Architecture and Results

We first applied each attribute group independently to the OLS model to compare their individual and collective explanatory power on housing prices. The five attribute groups' collective contribution using R^2 as the criterion are ranked as: location (0.678) > neighborhood (0.556) > objective streetscape attributes (0.382) > subjective streetscape scores (0.275) > structural attributes (0.188). All of them surpassed the required F-statistic test $p < 0.01$ criterion, which indicates the importance of the impacts of each of these groups. Furthermore, we ran the Moran's I test on OLS residuals and the results for each attribute group range from 0.42 to 0.65, indicating serious spatial autocorrelation. The results of LM tests also confirmed the presence of spatial lag and error, which implies the use of a spatial regression model, in our case the Kelejian and Prucha model.

Then, a baseline model (Model 0) was set using all macro-level factors, including structural, locational, and neighborhood attribute groups. VIF was used to identify correlation issues. Features that were of less importance using GI [56] and with multicollinearity ($VIF > 10$) were deleted. For instance, the distance to school was correlated with accessibility measures but had lower GI and was thus removed from the model. This indicates that access to mixed POI is more important than access to one single type of facility [39]. This model was able to explain 78.3% price variance.

Based on this, accounting for micro-level attributes, we added five subjective perceptions (Model 1) and ten objective feature indices (Model 2) separately for comparison (Table 2). GI and VIF were calculated similarly as in Model 0 to ensure the soundness of the test. For Model 2, the VIF of sky was larger than the threshold and was thus removed. The test on the three models revealed that all three models do not report strong multicollinearity issues.

Table 2. Regression performance and diagnosis for all models.

Model Attributes	Model 0 (Baseline)	Model 1 (Subjective Perceptions)	Model 2 (Objective View Indices)	Model 3 (Subjective Perceptions)	Model 4 (Objective View Indices)
Regression Model	OLS	OLS	OLS	Kelejian and Prucha's Model	Kelejian and Prucha's Model
Adjusted R^2 (Pseudo R^2)	0.739	0.744	0.759	(0.791)	(0.739)
Moran's I on Residual (p-value)	0.2034 ***	0.1857 ***	0.1678 ***	−0.0021 (0.1)	0.0007 (0.59)
Robust LM (lag)	283.859 ***	428.719 ***	4038.061 ***		
Robust LM (error)	16,589.358 ***	13,433.823 ***	14,780.848 ***		

Note: p values are shown in parentheses; *** < 0.01.

Model 1 and Model 2 both outperformed the Baseline Model (Model 0) regarding R^2 (Table 2), which indicates its effectiveness in integrating micro-level street factors to enhance the housing price prediction ability. We also found that the objective view indices are slightly more robust than subjective perceptions, which is consistent with previous research [61]. Since the spatial autocorrelation issue was detected after the tests, the spatial

HPM model was used. The regression result from the five models (including both OLS and Kelejian and Prucha's model) is shown in Appendix B.

Based on the spatial model results, considering spatial error and spatial lag interaction effects together, both Model 3 and Model 4 yield solid goodness-of-fit compared to the OLS. Since the Moran's I test on residuals became statistically insignificant for Models 3 and 4, it proves that the spatial HPM model tackled the autocorrelation issues. It is worth noting that accounting for spatial interaction does not change the sign nor the significance in parameter estimates for most of our investigated variables. Moreover, their magnitude of coefficients remained relatively stable when comparing the OLS and Kelejian and Prucha model. On the one hand, within the objective feature group, only very few indicators, i.e., tree, sign, and sidewalk, in the spatial model showed smaller coefficients. This suggests that there is mild bias in parameter estimation for these three indicators if we solely depend on the OLS. On the other hand, all the objective variables showed sufficient significance, while four of the five subjective perceptions retained the significance in the spatial model except for the imageability. Previous studies found a similar result of imageability becoming insignificant using the spatial model [55]. The training accuracy could cause this due to the representativity issue of the selected training SVI as well as a relatively lower consensus on evaluating this complex and nebulous concept.

4. Discussion

It is worth noting that our primary focus was to examine the impacts of micro-level perceived characteristics and not the discrepancies between HPM and the following spatial HPM model. Given that the sign, magnitude, and significance of the OLS coefficients largely resonate with the Kelejian and Prucha Model, and that the interpretation of the spatial models could be complicated and sometimes misleading, the discussion of the results would go beyond the scope of this research. Thus, the following discussion sections will be based on results from the OLS Models 1 and 2.

4.1. Effects of Conventional Macro-Level Attributes

Although our research focuses on uncovering the impact of micro-level indicators, we echoed the impacts of macro-level variables on housing prices [30,59,62]. All four neighborhood attributes are within the top ten factors, which is in agreement with prior research [30,62]. Accessibility to the metro and high school as well as job density and service density (A2MTR, A2SCH, DENWRK, DENSRV, respectively) are all positively associated with prices, and three of them ranked in the top five by GI score. Our result also further strengthened the impact of accessibility to good high schools and the metro. The 380 RMB/m² and 131 RMB/m² price increase for a 1 km and 5 km radius revealed a high price premium for a 10% increase regarding the schools and metro service in Shanghai.

Second, the GI score shows that location attributes play a vital role in affecting the housing prices; the distance to the CDB (D2CBD) ranks second among all the variables. This further supports previous research findings [59]. Specifically, when the distance to the CBD is 10% longer, the housing prices will decrease by around 1.1%. This is intuitive as more amenities and services concentrate around the central business district areas and provide more convenience and opportunities for social life in the city, which is a value cherished in daily lives, and is thus reflected in a higher living cost.

Third, structural attributes consistently show a positive sign in affecting prices [30]. More intricate decoration or design style, whether a room faces south to receive ample sun, and elevator facilities are all related to housing price increases ranging from 0.85% (488 RMB/m²) to 3.5% (1969 RMB/m²).

4.2. Effects of Micro-Level Attributes (Subjective vs. Objective)

Our HPM study proves that housing prices can be significantly affected by micro-level human-perceived features, which were largely neglected in prior research. The model further reveals the large variances between subjective perceptions and objective features.

First, using the GI score (Table 3) to rank the attributes, the objective features overall explain better than their subjective perceptions. For example, the road view index ranks the highest among all the micro-level variables, which is also consistent with the result from Table 3, which shows that the objective features collectively have higher strength than their subjective counterparts. We expanded the previous understanding of the impacts of objective street elements, including tree and building elements [34,37], and concluded that many other features (e.g., wall and sidewalk) can also provide direct impacts on the visual quality of the streetscape, which influences neighborhood preferences and in turn housing prices.

Second, if using the standardized coefficient (Table 3) to scrutinize the influences of both types of perceived attributes, it reveals that individual subjective perception exhibits a higher strength than the objective features. For example, safety is the strongest while it is closely followed by human scale and complexity. However, it is worth noting that all the objective features were out of the top 15 variables. Moreover, less discussed subjective perceptual qualities such as human scale show better strength than the agreed impact of visual enclosure [9]. These findings are further supported by the Gini importance calculations (higher values).

Third, Figure 5b provides a more direct reflection of the impact of street perceptual quality by increasing the scores by 10% to show the change in the housing prices. As previous studies have indicated the preference of openness by many homebuyers [37,45,49], it is not surprising that the objective view of building and sidewalk and the subjective perception of enclosure and human scale all negatively relate to the housing prices as they are opposite concepts to openness. As the statistics show, the effect of a 10% increase in subjective scores is an expected much higher rise in housing prices than the objective features. In particular, the complexity ranks the highest among subjective variables and foresees a 1232 RMB/m² value increase, while the highest objective variable, i.e., road, only increases the price by 206 RMB/m². It also shows some mixed results compared to prior research. The tree view increase in our model reveals less impact (129 RMB/m² vs. 710 RMB/m²) than the green view index with the 10% score increase [37]. We suggest that the discrepancy is partially because the tree index only extracts the quantity of trees from the scene while the green view index jointly analyzes the quantity of other vegetation including understory plants in the lower halves of the image [63,64]. Scholars have also argued that the impact of perceived greenness might also be related to quality as well as quantity [17]. These findings reveal that subjective perceptions impact housing price changes in a higher magnitude partly because they capture more overall information that can better represent the comprehensive cognition process of people, which cannot be explained well by each objectively measured feature view index.

4.3. Comparing Subjective and Objective Micro-Level Attributes

On the one hand, the statistical strength of the overall subjective perception compared to the objective features relies on its inclusiveness in encompassing the psychological and socio-demographic aspects of the general public in a tangible way. The subjective evaluation of a street scene enables viewers to judge the overall appearance and quality through refined details and subtle elements, which is hard to quantify simply by extracting pixels of an objective feature. Objectively measured perceived elements may oversimplify the power of architectural and landscape architectural design in placemaking. For example, the human scale perception is defined by how people perceive the texture and detail at a pedestrian walking speed [42]. For this type of experiential quality that is a more comprehensive perception, it is of great advantage to apply a subjective measurement process compared to objective features to explain it. Previously, scholars have tried to use complex mathematical formulas based on view indices and operative definitions to calculate directly from the street scenes [45]. Nevertheless, their statistical strength cannot reach the same standard as their subjective counterparts [38,39]. The sensorial experience often provides a more direct trigger for emotions and thus affects human behaviors. Scholars

have also reported the subjective urban design perceptions to have mediating effects for psychological perceptions [65]. Therefore, the psychological mechanism of how people read the visual landscape needs further attention. Conversely, for more straightforward perceptions such as enclosure, objective features could possibly replace the subjective perceptions and achieve higher accuracy.

Table 3. OLS regression model for comparison and interpretation.

OLS Regression Variables		Multilinearity and Feature Importance			Interpretation (Based on Unstandardized Coef.)		
Variables	VIF	Gini Score	Std. Coef.	Ranking (Std. Coef.)	Delta X	% Price Change	RMB/m ² Change
Structural Attributes							
FLAREA	5.5	0.016	−0.038	20	1 unit	−0.01	−¥8
BEDRM	2.9	0.007	−0.022	23	1 unit	−0.42	−¥240
BATH	3.1	0.005	0.080	13	1 unit	2.40	¥1378
CSTRYR	2.3	0.030	0.125	4	1 unit	0.22	¥124
ELEVTR	3.7	0.047	0.099	8	T/F	3.43	¥1969
HGHT	1.1	0.006	−0.049	16	T/F	−1.75	−¥1001
TWR_SLB	1.3	0.009	−0.108	6	T/F	−6.48	−¥3715
STH_NTH	1	0.003	0.020	24	T/F	0.85	¥488
DECOR	1.1	0.007	0.085	10	T/F	2.91	¥1666
Neighborhood/Location Attributes							
LN (D2CBD)	3.7	0.253	−0.415	1	10%	−1.12	−¥642
LN (DENWRK)	1.2	0.014	0.005	27	10%	0.01	¥3
LN (DENSrv)	2	0.036	0.040	18	10%	0.05	¥26
LN (A2MTR)	2.2	0.076	0.137	3	10%	0.23	¥131
LN (A2SCH)	5	0.311	0.410	2	10%	0.66	¥380
Subjective Streetscape Perception Attributes							
LN (S1_ENCLS)	9.3	0.007	−0.081	11	10%	−0.90	−¥514
LN (S2_HMSCL)	5	0.008	−0.102	7	10%	−1.44	−¥824
LN (S3_CMPLX)	3.8	0.007	0.095	9	10%	2.15	¥1232
LN (S4_IMGBL)	3.5	0.006	0.080	12	10%	1.16	¥665
LN (S5_SAFTY)	6.6	0.005	0.117	5	10%	1.34	¥768
Objective Streetscape Feature Attributes							
LN (O1_SKY)	/	/	/	/	/	/	/
LN (O2_TREE)	2.1	0.005	0.057	15	10%	0.23	¥129
LN (O3_BLDG)	3.4	0.006	−0.038	19	10%	−0.12	−¥68
LN (O4_PRSN)	1.3	0.007	0.017	25	10%	0.01	¥7
LN (O5_CAR)	1.3	0.006	0.045	17	10%	0.07	¥39
LN (O6_ROAD)	2	0.008	0.071	14	10%	0.36	¥206
LN (O7_SDWK)	1.5	0.006	−0.033	21	10%	−0.08	−¥47
LN (O8_FENC)	1.3	0.007	0.022	22	10%	0.03	¥16
LN (O9_WALL)	1.4	0.008	−0.002	28	10%	0.00	−¥1
LN (O10_SIGN)	1.2	0.006	0.009	26	10%	0.01	¥5

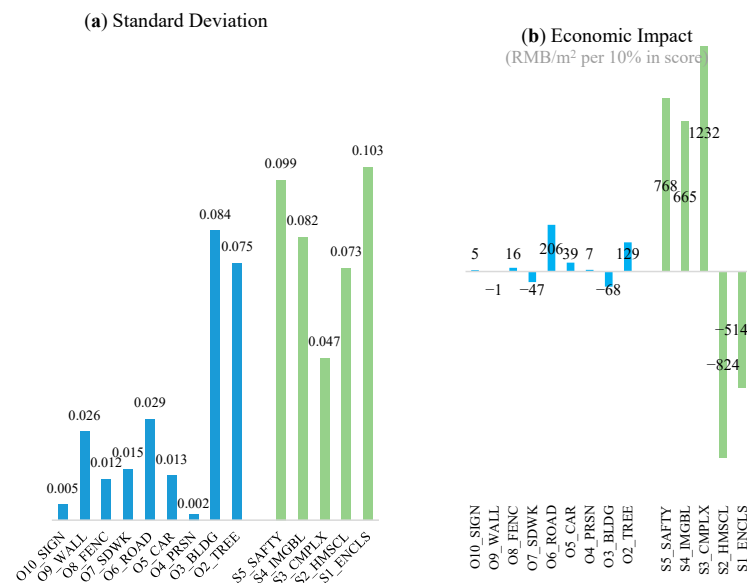


Figure 5. (a) Standard deviation of micro-level attributes. (b) Economic implications on the property values (RMB/m²) if increasing scores by 10%.

On the other hand, objective features can still provide benefits to support subjective perceptions. First, perceptions are multifaceted and complex, making it challenging to use and exhaust all the perceptions for evaluation. Scholars have tried to measure over fifty qualities but the majority of them failed to give good results in previous empirical studies [42]. Nevertheless, through algorithms it is relatively easier to extract all the objective features, and their presence in SVIs is more stable and prevailing; thus, they can be explicitly modeled in the research. Most of the features we extracted have already been used in various SVI studies [37,40,45]. Second, another advantage is that objective view indices barely have any multicollinearity issues compared to subjective perceptions [23,39]. Previous scholars have tried to use more complicated statistical models including PCA to cut the dimensionality, but this creates more composite multisubjective perceptions which might limit the application of these perceptions and it may also distort its original interpretation of the concept [38]. Furthermore, the more perceptual qualities we want to measure, the more training datasets need to be collected, which will be more time- and cost-consuming. Thus, future research should consider combining view indices to complement the subjective perceptions that are deleted due to multicollinearity issues and increase the overall explanatory strength of the model.

Therefore, when choosing between subjective perception and objective attributes in predicting certain factors, these variables should be carefully chosen to avoid multicollinearity issues and concept overlap. Subjective and objective attributes both have their advantages, and it should be further investigated how they can complement each other and attain good results.

4.4. Policy Implications

Our research provides useful tools which can effectively predict housing prices, and our findings also reveal important policy and urban planning implications. First, the investigation discloses the value of the micro-level neighborhood environment attributes, which was neglected in prior studies. Developers should work with government agencies to improve the micro-level perceived environment quality by accommodating amenities in the public realm. For example, around residential locations, street furniture needs to be visually perceptible and accessible to pedestrians, which can help elevate the neighborhood street vitality. Second, the study can help policymakers to make decisions on the overall needs of citizens and facilitate a sustainable development plan for the city. For instance, real estate developers enjoy a price premium because of the better micro-level perceived

qualities of the street environment while the implementation and maintenance costs are allocated from the government's revenue. Upon agreement, street environment tax could be levied on developers based on the assessment to compensate this cost [36]. On the other hand, developers should receive necessary policy incentives if they fully engage in the public street environment planning process and take over its implementation cost. Finally, measuring the perceptions themselves can provide meaningful guidance on the citywide mapping using various attributes to potentially understand the social equity issue [38] and help efficiently identify neighborhood locations that need prioritized development or renovation [45].

5. Limitations and Future Directions

We would also like to point out several limitations of our study. First, our research aims to study the correlation between housing prices (neighborhood revealed preference) and micro-level subjective and objective urban perception attributes using the HPM procedure. This type of research design limits the opportunity to define its potential causal relationship because of its cross-sectional nature. The confounding factor with both the housing prices and street environment were not discussed either, such as a decisive policy for beautifying the neighborhood. However, it is meaningful to define the causal relationship between the housing prices and built environment factors as this can provide necessary suggestions on policy for city staff and more in-depth and profound empirical results that are relatively scarce in the urban studies literature [24]. Apart from acquiring longitudinal time-series data, the time matching between two datasets should also be carefully examined. It is worth noting that the HPM can possibly misrepresent the monetized value of an urban public space if the estimation of the impacts of certain attributes are biased, and we should realize that the value also includes social and ecological value and those implied assets [66].

Second, thermal comfort experienced at the street level provides various benefits and is related to health and social equity issues [67]. Similarly, other types of temporal experience such as noise level should also be considered as these factors affect housing choices. Moreover, the quality of the landscape features, such as whether the pavement is degraded, were not considered, and the architecture façade style can be further taken into account using deep learning algorithms [68]. Scholars have also argued that because of the difference in traffic volume among each type of street (e.g., main street, residential street), the streets' hierarchy should be considered [69]. A future HPM can include these factors as additional variables to provide more comprehensive inferences.

Third, more macro-level attributes can be added in future studies or examined in a different approach. For example, the number of each type of POI can be used as an individual variable in the HPM model to further scrutinize their impacts and how they compare with micro-level perceptions. High-quality primary schools or middle schools can also be added in the future.

Fourth, due to SVI data limitation, we could only obtain general street views through Baidu Street View Static API in our study. We acknowledge that both street views around the neighborhood and inside would affect housing prices. However, because most Chinese communities are gated, views inside are often not downloadable [11,36]. Moreover, the windowed views inside the room which are specific to each property can also be influential in affecting housing prices because good views can provide psychologically restorative effects. A recent study by Li et al. [70] used a high-resolution city information model combined with the deep transfer learning method to automatically assess windowed views for high-rise contexts in Hong Kong. This method can be useful in studying its impact on property values in future studies.

Fifth, regarding the streetscape perceptual quality, more advanced statistical models such as PCA can be applied in the future to jointly analyze its impacts on predicting housing prices [22,38]. Furthermore, following prior studies, the prediction on subjective perception scores in our research completely relies on high-level features (i.e., streetscapes) as explanatory variables to ensure interpretability. The prediction accuracy can be improved

by integrating low-level features such as edges or hues [54,63]. The perception training data could also come from non-experts (such as potential homebuyers) when evaluating more sensorial perceptions similar to the MIT Place Pulse dataset [71], which may provide interesting or even contrasting opinions. Given the data collection and analysis effectiveness, we plan to apply this framework to study other metropolitan cities in comparison to Shanghai, which would allow us to gain more insights.

Last, it is quite challenging to interpret these micro-level perceived attributes and provide an actionable recommendation. For example, further thought should be given to how to strategically increase 10% of the street complexity perception and what its implications for design implementation would be. Future research can provide a more in-depth study on the relationship between design features and perceived street quality, further offering evidence-based design suggestions, for instance, the implications for floor area ratio adjustment [72]. It will provide more insights to inform the urban design guidelines that can ultimately facilitate an engaging street interface that connects people with social life.

6. Conclusions

The prevailing studies have well understood how conventional macro-level attributes affect housing prices. While emerging studies have increasingly applied CV algorithms and SVIs to investigate the micro-level neighborhood street quality, these measures mainly account for a single or a few visual features quantified objectively. Using macro-level attribute groups as control variables, we tremendously increased the comprehensiveness of the objective streetscape features studied in our model. We also provided interpretations on previously neglected subjective perceptions of the urban design qualities using a classical theoretical framework [42] with the help of new analytical tools and crowdsourcing surveys to effectively analyze how property values are informed by the way people perceive their neighborhood environment.

Using our high-throughput framework, the study quantified five subjective perceptions and ten objective features from SVI datasets of Shanghai. Employing HPM and spatial HPM methods, our study revealed that micro-level attributes including both subjective perceptions and objective features explain more variances than conventional structural attributes. Moreover, although objective measures have higher collective strengths over subjective perceptions, individual subjective perceptions have stronger explanatory power than individual objective features, implying that subjective perceptions can more comprehensively capture unexplained built environment factors. On the other hand, objective measures can further complement subjective counterparts since they are less affected by multicollinearity issues.

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Informed Consent Statement: Written informed consent was waived as the analyzed dataset was properly anonymized, so no participant can be identified.

Data Availability Statement: Data is available upon reasonable request.

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Appendix A

Table A1. Descriptive statistics of all HPM variables.

Variables	Description		Count	Mean	Std. Dev.	Min	Max	Data Source
PRICE	Transaction price (RMB/m ²)		40,159	57,349	21,683	10,400	250,813	Lianjia.com (accessed on 20 August 2022)
STRUCTURAL ATTRIBUTES								
FLAREA	Total floor area (m ²)		40,159	85	43	15	588	Web scraping from Lianjia.com (accessed on 20 August 2022)
BEDRM	Number of bedrooms		40,159	2.1	0.8	1	8	
LIVRM	Number of living rooms		40,159	1.4	0.6	0	5	
KITCH	Number of kitchens		40,159	1.0	0.2	0	5	
BATH	Number of bathrooms		40,159	1.2	0.5	0	7	
TTLFLR	Total floors of the building		40,159	11.0	7.9	1	62	
CSTRYR	Construction year of the building		40,159	1998	9.4	1912	2019	
	Description	Values	Count	%	Ave. Price (¥/m ²)	Ave. Area (m ²)		Data Source
HGHT	On which floor in the building is the unit located?	0: Low /Mid 1: High	23,075 17,084	57.5% 42.5%	59,021 55,092	90 79		Web scraping from Lianjia.com (accessed on 20 August 2022)
LAYT	The layout of the unit	0: Duplex 1: Flat	1632 38,527	4.1% 95.9%	58,108 57,317	154 82		
TWR_SLB	The shape of the building	0: Slab 1: Tower	36,591 3568	91.1% 8.9%	56,346 66,706	85 88		
STH_NTH	Is the unit south-facing?	0: Else 1: South	7993 32,166	19.9% 80.1%	56,110 57,657	94 83		
STRC	The structure of the building	0: Brick 1: Steel	18,003 22,156	44.9% 55.2%	53,060 60,819	61 105		
DECOR	Interior decoration quality of the unit	0: Simple 1: Refined	19,300 20,859	48.10% 51.90%	53,680 61,322	74 96		
ELEVTR	Is an elevator available?	0: No 1: Yes	24,106 16,053	60.0% 40.0%	52,764 64,235	69 110		
LOCATION ATTRIBUTES								
			Count	Mean	Std. Dev.	Min	Max	Data Source
D2SCBD	Network distance (km) to the district geometric center		40,159	4.77	3.04	0.02	16.29	Computed in QGIS, with Shanghai (2018) shapefile
D2CBD	Network distance (km) to the city center (the Bund)		40,159	12.62	7.48	0.03	35.11	
	Description	Values	Count	%	Ave. Price (¥/m ²)	Ave. Area (m ²)		Data Source
RING_X	Within which ring road is the unit located? X stands for the ring index.	1: Inner ring 2: Middle ring 3: Outer ring 4: Outskirt ring	9290 9835 8742 12,292	23.1% 24.5% 21.8% 30.6%	81,151 63,057 52,356 38,345	88 79 81 92		Web scraping from Lianjia.com (accessed on 20 August 2022)
CTY_XX	In which district is the unit located? XX stands for the district name. In the final models, Baoshan district (i.e., CTY_BS) was the base group.	BS: Baoshan	3390	8.4%	44,159	81		
		CN: Changning	2400	6.0%	70,051	83		
		FX: Fengxian	992	2.5%	24,524	95		
		HK: Hongkou	1513	3.8%	66,210	80		
		HP: Huangpu	1267	3.2%	92,725	103		
		JA: Jin'an	964	2.4%	95,101	90		
		JD: Jiading	1662	4.1%	37,527	87		
		MH: Minhang	4806	12.0%	49,479	91		
		PD: Pudong	9389	23.4%	57,590	87		
		PT: Putuo	2941	7.3%	58,412	76		
		QP: Qingpu	678	1.7%	30,976	94		
		JS: Jinshan	2201	5.5%	36,432	100		
		XH: Xuhui	3060	7.6%	74,879	79		
		YP: Yangpu	3091	7.7%	62,677	72		
		ZB: Zhabei	1805	4.5%	63,647	79		

Table A1. Cont.

NEIGHBORHOOD ATTRIBUTES		Count	Mean	Std. Dev.	Min	Max	Data Source
DENSRV	Density of living amenities and services (thousand/km ²)	40,159	0.115	0.187	0	3.5	Dazhongdianping.com (accessed on 20 August 2022), density calculated in ArcGIS
DENWRK	Density of office (thousand/km ²)	40,159	9.5	22.4	0	573.5	
D2MTR	Distance to metro (km)	40,159	0.8	0.7	0.01	7.8	Distances calculated in Python
A2MTR	Accessibility to metro	40,159	5.7	6.8	0	46.0	
D2SCH	Distance to school (km)	40,159	2.7	2.3	0.02	11.9	
A2SCH	Accessibility to school	40,159	7.0	7.0	0	29.0	
SUBJECTIVE STREETSCAPE ATTRIBUTES							
S1_GREEN	Subjectively perceived greenness	40,159	0.8	0.0	0.4	0.9	Predicted by ML models with physical feature view indices extracted from Baidu SVIs
S2_WALKB	Subjectively perceived walkability	40,159	0.6	0.1	0.4	0.8	
S3_SAFTY	Subjectively perceived safety	40,159	0.7	0.1	0.3	1.0	
S4_IMBLT	Subjectively perceived imageability	40,159	0.7	0.1	0.3	0.9	
S5_ENCLS	Subjectively perceived enclosure	40,159	0.7	0.1	0.3	0.9	
S6_CMPLX	Subjectively perceived complexity	40,159	0.6	0.0	0.5	0.9	
OBJECTIVE STREETSCAPE ATTRIBUTES							
O1_GREEN	Objectively derived greenness	40,159	0.4	0.1	0.0	0.8	Derived scores by recombining selected physical feature view indices
O2_WALKB	Objectively derived walkability	40,159	0.6	0.1	0.2	0.7	
O3_SAFTY	Objectively derived safety	40,159	0.4	0.1	0.1	0.7	
O4_IMBLT	Objectively derived imageability	40,159	0.6	0.1	0.0	0.9	
O5_ENCLS	Objectively derived enclosure	40,159	0.6	0.0	0.1	0.7	
O6_CMPLX	Objectively derived complexity	40,159	0.3	0.1	0.0	0.6	

Appendix B

Table A2. OLS and Kelejian and Prucha model regression results.

	Model 0 (Baseline)		Model 1 (Subjective Perceptions)		Model 2 (Objective View Indices)		Model 3 (Subjective Perceptions)		Model 4 (Objective View Indices)	
Model Type	OLS		OLS		OLS		Kelejian & Prucha		Kelejian & Prucha	
Variables	Coef.	<i>p</i> > <i>t</i>	Coef.	<i>p</i> > <i>t</i>	Coef.	<i>p</i> > <i>t</i>	Coef.	<i>p</i> > <i>t</i>	Coef.	<i>p</i> > <i>t</i>
Constant	0.4041		0.4298		0.6830		0.2555	***	1.0317	***
Structural Attributes										
FLAREA	−0.0001	***	−0.0001	**	−0.0001	**	0.0002	***	0.0004	***
BEDRM	−0.0040	*	−0.0037		−0.0047	**	−0.0053	***	−0.0018	***
BATH	0.0244	***	0.0238	***	0.0243	***	0.0252	***	−0.0214	***
CSTRYR	0.0022	***	0.0022	***	0.0021	***	−0.0002	**	−0.0010	***
ELEVTR	0.0350	***	0.0348	***	0.0338	***	−0.0310	***	0.0196	***
HGHT	−0.0166	***	−0.0172	***	−0.0177	***	0.0003		0.0035	
TWR_SLB	−0.0670	***	−0.0650	***	−0.0645	***	−0.0163	***	−0.0755	***
STH_NTH	0.0084	***	0.0085	***	0.0085	***	−0.0124	***	0.0530	***
DECOR	0.0295	***	0.0293	***	0.0289	***	0.0161	***	0.0579	***
Neighborhood/Location Attributes										
LN (D2CBD)	−0.1143	***	−0.1102	***	−0.1136	***	−0.0008	**	0.0175	***
LN (DENWRK)	0.0008	**	0.0006	**	0.0005	*	−0.0012	***	0.0116	***
LN (DENSrv)	0.0047	***	0.0048	***	0.0043	***	−0.00027	***	−0.0129	***
LN (A2MTR)	0.0216	***	0.0234	***	0.0222	***	0.0031	***	0.0065	***
LN (A2SCH)	0.0646	***	0.0657	***	0.0670	***	0.0074	***	−0.0030	***
Subjective Streetscape Perception Attributes										
LN (S1_ENCLS)	/		−0.0896	***	/		−0.0619	***	/	
LN (S2_HMSCL)	/		−0.1438	***	/		−0.1293	**	/	
LN (S3_CMPLX)	/		0.2147	***	/		0.1415	***	/	
LN (S4_IMGBL)	/		0.1160	***	/		0.0286		/	
LN (S4_SAFTY)	/		0.1339	***	/		0.1393	***	/	

Table A2. Cont.

	Model 0 (Baseline)	Model 1 (Subjective Perceptions)	Model 2 (Objective View Indices)	Model 3 (Subjective Perceptions)	Model 4 (Objective View Indices)
Model Type	OLS	OLS	OLS	Kelejian & Prucha	Kelejian & Prucha
Objective Streetscape Feature Attributes					
LN (O1_SKY)	/	/	/	/	/
LN (O2_TREE)	/	/	0.0225 ***	/	0.0455 ***
LN (O3_BLDG)	/	/	−0.0118 ***	/	−0.034 ***
LN (O4_PRSN)	/	/	0.0012 ***	/	0.0035 ***
LN (O5_CAR)	/	/	0.0068 ***	/	0.0108 ***
LN (O6_ROAD)	/	/	0.0360 ***	/	0.0218 ***
LN (O7_SDWK)	/	/	−0.0083 ***	/	−0.0211 ***
LN (O8_FENC)	/	/	0.0028 ***	/	0.0050 ***
LN (O9_WALL)	/	/	−0.0002 ***	/	−0.0012 ***
LN (O10_SIGN)	/	/	0.0009 ***	/	0.0105 ***
Spatial Lag/Error					
LAG(LN(PRICE))		/		0.0201 ***	0.0066 ***
LAMBDA		/		−0.2361 ***	0.6432 ***

Note: ***, **, and * indicate significance level of 1%, 5%, and 10%, respectively.

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